

Problem 1: A motor, power and torque

$$(1) \text{ Shear Stress in a Shaft: } \tau = \frac{Tc}{J}$$

τ	Shear Stress in a Shaft	N/mm ² aka MPa
T	Torque	N*m (convert to N*mm)
c	Radius	mm
J	Polar MOI	mm ⁴

$$\text{Polar MOI Solid Shaft: } J = \frac{\pi}{2} (r^4)$$

$$\text{Polar MOI Hollow Shaft: } J = \frac{\pi}{2} (r_{\text{outer}}^4 - r_{\text{inner}}^4)$$

Shear Stress Steps

1. Find ω in rad/s
2. Find $T_a, T_b, \dots$ using $P = T\omega$ (Usually N*m, change to mm by *1000)
3. Find J, use it to calculate $\tau_{ab}, \tau_{bc}, \dots$

Problem 2: Angle of twist

$$(2) \text{ Angle of Twist (rad): } \phi = \frac{TL}{JG}$$

ϕ	Angle of Twist	rads, dimensionless
T	Torque	N*m (convert!)
L	Length of Section	mm
J	Polar MOI	mm ⁴
G	Shear Modulus of Elasticity	GPa, convert to MPa

$$(3) \text{ Power: } P = T \cdot \omega$$

P	Power	Watt or ft/lb/s
T	Torque	N*m/ft*lb (convert!)
ω	Length of Section	mm

Power (P) = Torque (T) · Rotational Speed (ω)		
SI Metric Units		
$P = T \cdot \omega$	P = Power (watt) T = Torque (N·m) ω = Rotational Speed (rad/s)	
$P = (0.1047) T \cdot \omega$	P = Power (watt) T = Torque (N·m) ω = Rotational Speed (rpm)	
English Units		
$P = T \cdot \omega$	P = Power (ft·lb/s) T = Torque (ft·lb) ω = Rotational Speed (rad/s)	
$P = \frac{T \cdot \omega}{5252}$	P = Power (hp) T = Torque (ft·lb) ω = Rotational Speed (rpm)	

$$1 \text{ rpm} = 2\pi/60 = 0.1047 \text{ rad/s}$$

$$1 \text{ hp} = 746 \text{ watt} = 550 \text{ ft·lb/s}$$

1. Find ω in rad/s (convert if necessary)
2. Power loss can be treated as $P(100\% - n\%)$
3. Find T_a, T_b using $P = T\omega$ (Usually N*m, to mm by *1000)
4. If ϕ is "F with respect to A," write $\phi_{AF} = \phi_{AB} + \phi_{BC} + \phi_{CD} + \phi_{DE}$
5. Using CCW is (+), CW (-), find T_{AB} to T_{DE} using cover up method
6. Calculate J
7. Calculate ϕ_{AF} as $\phi_{AB} + \phi_{BC} + \phi_{CD} + \phi_{DE}$

Problem 3: V and M diagram

1. Write order of the lines
2. Find EoE, define y reactions, change distributed loads to concentrated loads
3. Write discontinuity lines
4. Draw V (kN, lb), identify if (+) or (-) area, find area of each segment
5. Draw M (kn*m, lb*ft), identify concentrated moment and add by area of V

Problem 4: Absolute maximum bending normal and shear stress in a beam structure

$$(4) \text{ Bending Stress: } \sigma = \frac{My}{I}$$

σ	Bending Stress	N/mm ² aka MPa or kips/in ² aka ksi
M	Bending Moment	N*mm or kip*ft
y	Distance from Neutral Axis	mm or ft
I	MOI about the NA	mm ³ or ft ³

$$\text{MOI of Rectangular Area about Centroid: } I = \frac{1}{12} bh^3$$

$$\text{MOI about Non-Centroidal Axis: } I = \frac{1}{12} bh^3 + Ad^2$$

Bending Normal Stress Steps

1. Find Neutral Axis ($\bar{y}$). Break down beam into multiple components, vertical cuts, horizontal, label 1,2,3. Draw reference axis at bottom for $\bar{y}$ calcs
2. Apply table method for calculating centroids

	y (dist from centroid to reference axis)	A	yA
1			
2			
3			
4			

3. Calculate Centroid/Neutral Axis: $\bar{y} = \frac{\sum yA}{\sum A}$, label in FBD
4. Find MOI about Non-Centroidal Axis, where d is NA - y.
5. Find Tensile/Compression if Top/Bottom.

$$(5) \text{ Transverse Shear: } \tau = \frac{VQ}{IT}$$

τ	Transverse Shear	N/mm ² aka MPa or kips/in ² aka ksi
V	Shear Force	N or lb
Q	First Area MOI	mm ³ or ft ³
I	Second Area MOI	mm ⁴ or ft ⁴
T	Thickness of Portion	mm or ft

$$\text{First Area MOI (mm}^3\text{): } Q = \Sigma yA$$

Where y = Distance from Y axis, A = Area of Section

$$\text{Second Area MOI (mm}^4\text{): } I = \frac{1}{12}bh^3 + Ad^2$$

Where A = Area, d is distance from centroid to composite NA

Maximum Shear Steps:

Step 1: Find Neutral Axis

Step 2: Find I (global)

Step 3: Find Q at area under/on top of each point

Step 4: Find τ_A to τ_B using (5)

